

$ab = ba \rightarrow \text{group } \mathbb{Z} \times \mathbb{Z}$
 $\frac{S \vee S'}{\mathbb{Z} \times \mathbb{Z}}$
 $S \vee S'$
 $\frac{S \vee S'}{\mathbb{Z} \times \mathbb{Z}}$
 $T: \mathbb{Z} \times \mathbb{Z}$
 $T \circ \mathbb{Z} \times \mathbb{Z}$
 $f: \mathbb{Z} \times \mathbb{Z}$

[illegible][illegible]

Handwritten notes on graph paper:

- Left side: $\frac{1}{2} \frac{1}{2} \frac{1}{2}$
- Middle: $\frac{1}{2}$
- Right side: $\frac{1}{2}$